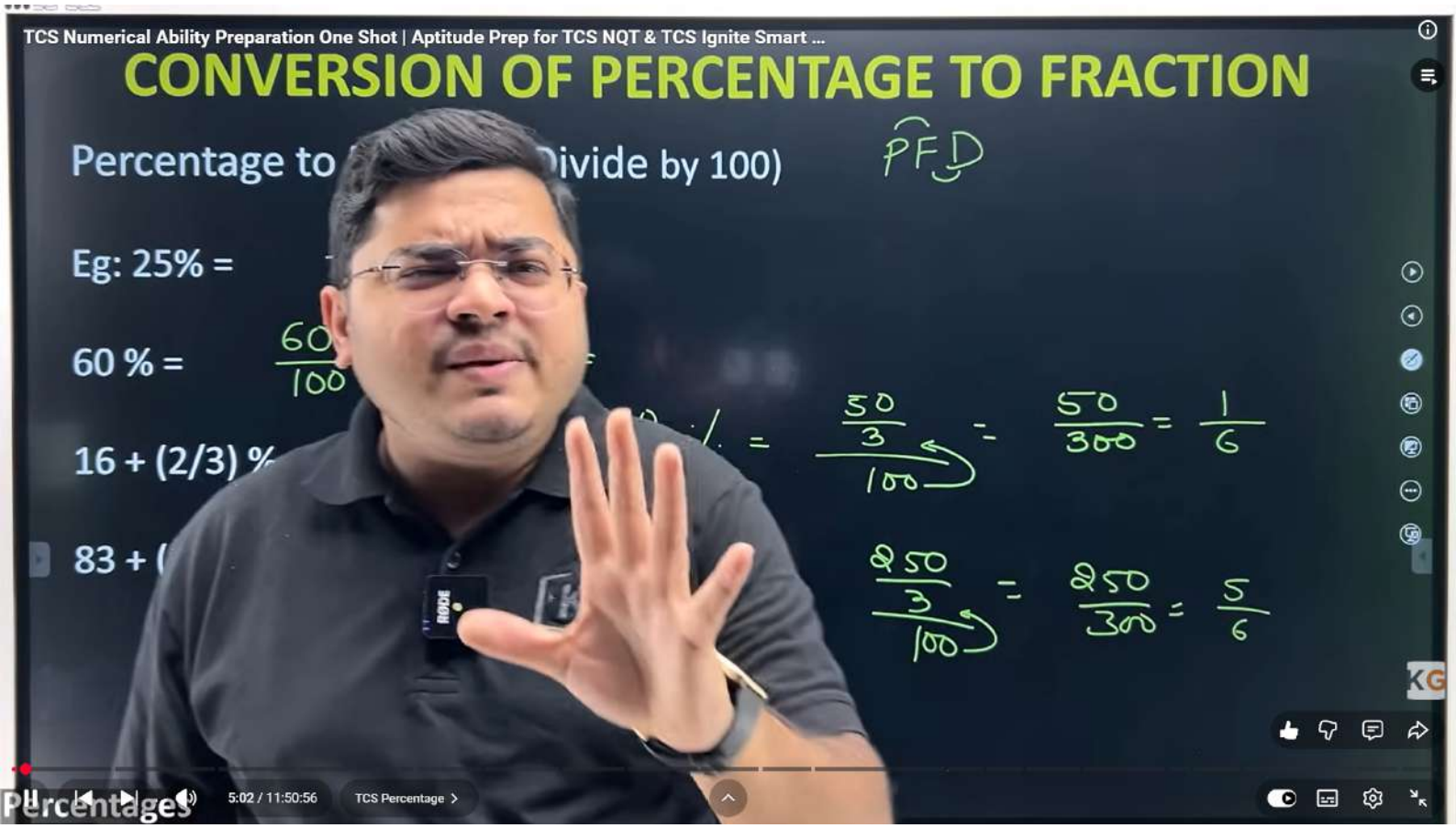
TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
CONVERSION OF PERCENTAGE TO FRACTION
Percentage to
ivide by 100)
PFD
Eg: 25% =
60 % =
60/100
16 + (2/3) %
83 +
50/3 / 100 = 50/300 = 1/6
250/3 / 100 = 250/300 = 5/6
Percentages
5:02 / 11:50:56
TCS Percentage

CONVERSION OF PERCENTAGE TO FRACTION

Percentage to Fraction (Divide by 100) PFD

Eg: 25% = $\frac{25}{100} = \frac{1}{4}$

% = $\frac{60}{100} = \frac{6}{10} = \frac{3}{5}$

16 + (2/3) % = $16\frac{2}{3} \Rightarrow \frac{50}{3}\% = \frac{\frac{50}{3}}{100} = \frac{50}{300} = \frac{1}{6}$

83 + (1/3) % = $83\frac{1}{3} \Rightarrow \frac{250}{3}\% = \frac{\frac{250}{3}}{100} = \frac{250}{300} = \frac{5}{6}$

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CONVERSION OF FRACTION TO PERCENTAGE

Fraction to Percentage (Multiply by 100)

FPM

3/4 = $\frac{3}{4} \times 100 = \frac{300}{4} = 75\%$

2/7 = $\frac{2}{7} \times 100 = \frac{200}{7} = 28.57\%$

$\boxed{\frac{2}{7} = 28.57\% \qquad \frac{1}{7} = 14.28\%}$

$$\begin{array}{r} 28.57 \\ 7\overline{)200} \\ -14 \\ \hline 60 \\ -56 \\ \hline 40 \\ -35 \\ \hline 50 \\ -49 \\ \hline 1 \end{array}$$

Percentages

8:05 / 11:50:56 TCS Percentage >

IMPORTANT CONCEPT

1. A number is increased by 20%

Original no = 100

new no = 120

$\frac{20}{100} = \frac{1}{5}$

Original no = 5

new no = 5 + 1 = 6

Q. A number is increased by 25%, then it becomes 600. The original number is __

Original no = y

$y + 25\%$ of $y = 600$

$y + \frac{25}{100} \times y = 600$

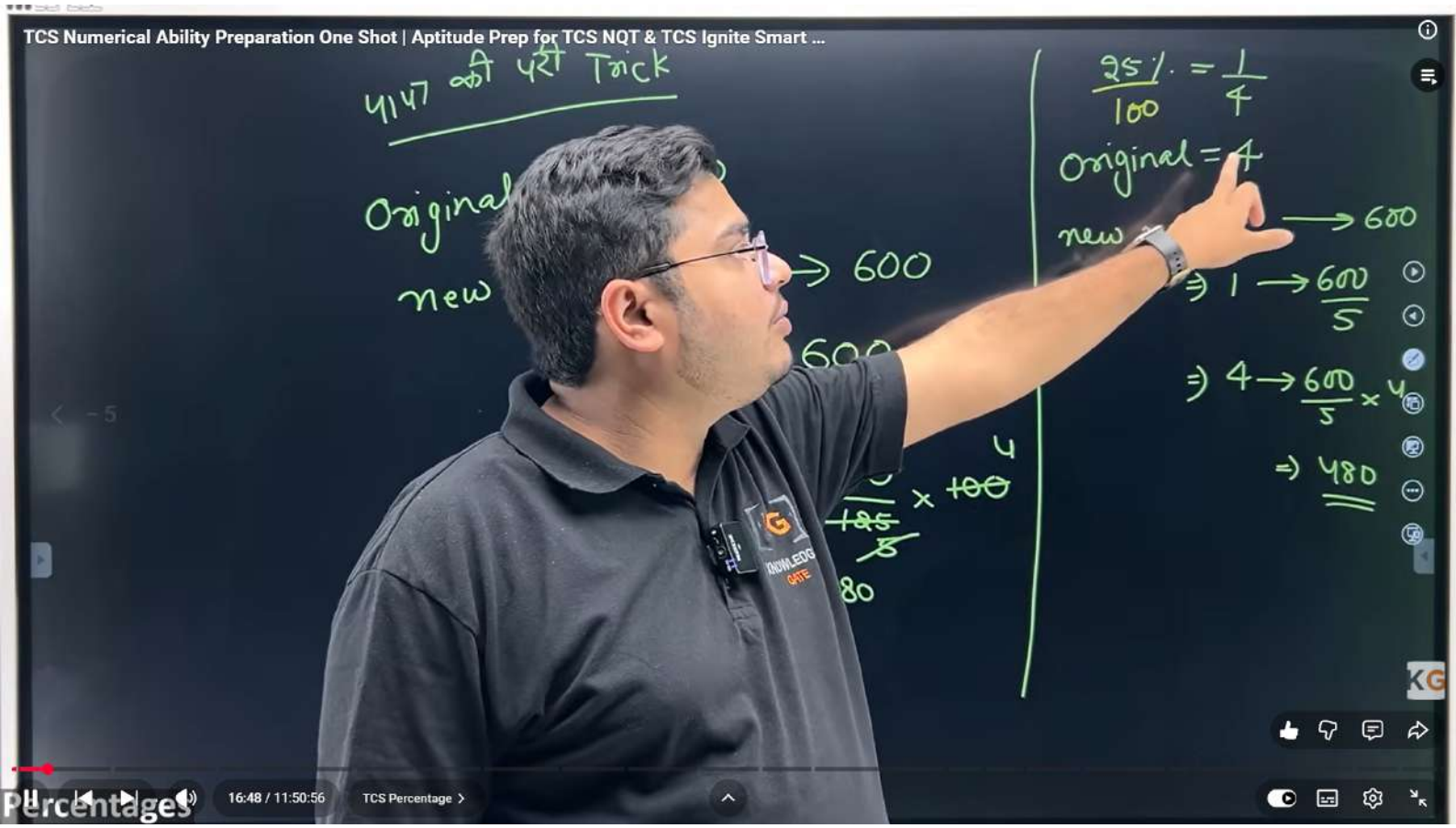
$y + \frac{y}{4} = 600$

$= 600$

120

$y = \ \times 4$

$\boxed{y = 480}$


TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
4147 की परी Trick
Original
new
→ 600
600
25% = 1/4
Original = 4
new
→ 600
⇒ 1 → 600/5
⇒ 4 → 600/5 × 4
⇒ 480
16:48 / 11:50:56
TCS Percentage
Percentages

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4147 की पूरी Trick

Original no = 100

new no = 125 → 600

$1 \rightarrow \frac{600}{125}$

$100 \rightarrow \frac{\cancel{600}^{120}}{\cancel{125}_{\cancel{5}}} \times \cancel{100}^{4}$

$\Rightarrow 480$

$\frac{25\%}{100} = \frac{1}{4}$

Original = 4

new no = $\overset{4+1}{5}$ → 600

$\Rightarrow 1 \rightarrow \frac{600}{5}$

$\Rightarrow 4 \rightarrow \frac{600}{5} \times 4$

$\Rightarrow 480$

17:04 / 11:50:56 TCS Percentage >

Percentages

Q. A number is decreased by 20%, then it becomes 120. The original number is __

$20\% = \frac{20}{100} = \frac{1}{5}$

orig = 5 → 5 —

new = 4 → 120

= 1 → 120/4 = 30

Percentages

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Q. A number is decreased by 20%, then it becomes 120. The original number is __

$20\% = \frac{20}{100} = \frac{1}{5}$

orig = 5 → 5 → 30×5 = 150 ✓

new = 4 → 120

= 1 → 120/4 = 30

18:04 / 11:50:56 TCS Percentage >

Percentages

Q. A number is decreased by 33 + 1/3 %, then it becomes 180. The original number is __

270

$33\frac{1}{3} = \frac{100}{3}\% = \frac{\frac{100}{3}}{100} \quad \frac{100}{300} = \frac{1}{3}$

orig = 3 $\xleftarrow{3 \to 270}$

new = 2 $\longrightarrow$ 180

1 $\longrightarrow$ 90

Q. The total income Yash Sir of 2017, 2018, 2019 is Rs. 364. Every year the salary increases by 20%. What is the income in 2017?

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A. Rs. 96
B. Rs. 100
C. Rs. 104
D. Rs. 108

Percentages

Q. The total income Yash Sir of 2017, 2018, 2019 is Rs. 364. Every year the salary increases by 20%. What is the me in 2017?

96

00

M2

let 2017 → Rs. 100

2018 → 100 + 20% of 100

→ Rs. 120

2019 → 120 + 20% of 120

→ 120 + 24

→ Rs. 144

100 + 120 + 144

= Rs. 364

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Q (VK). The total income of Eesha in the years 2003, 2004, 2005 was $36,400. Her income increased by 20% each year. What was her income in 2005?

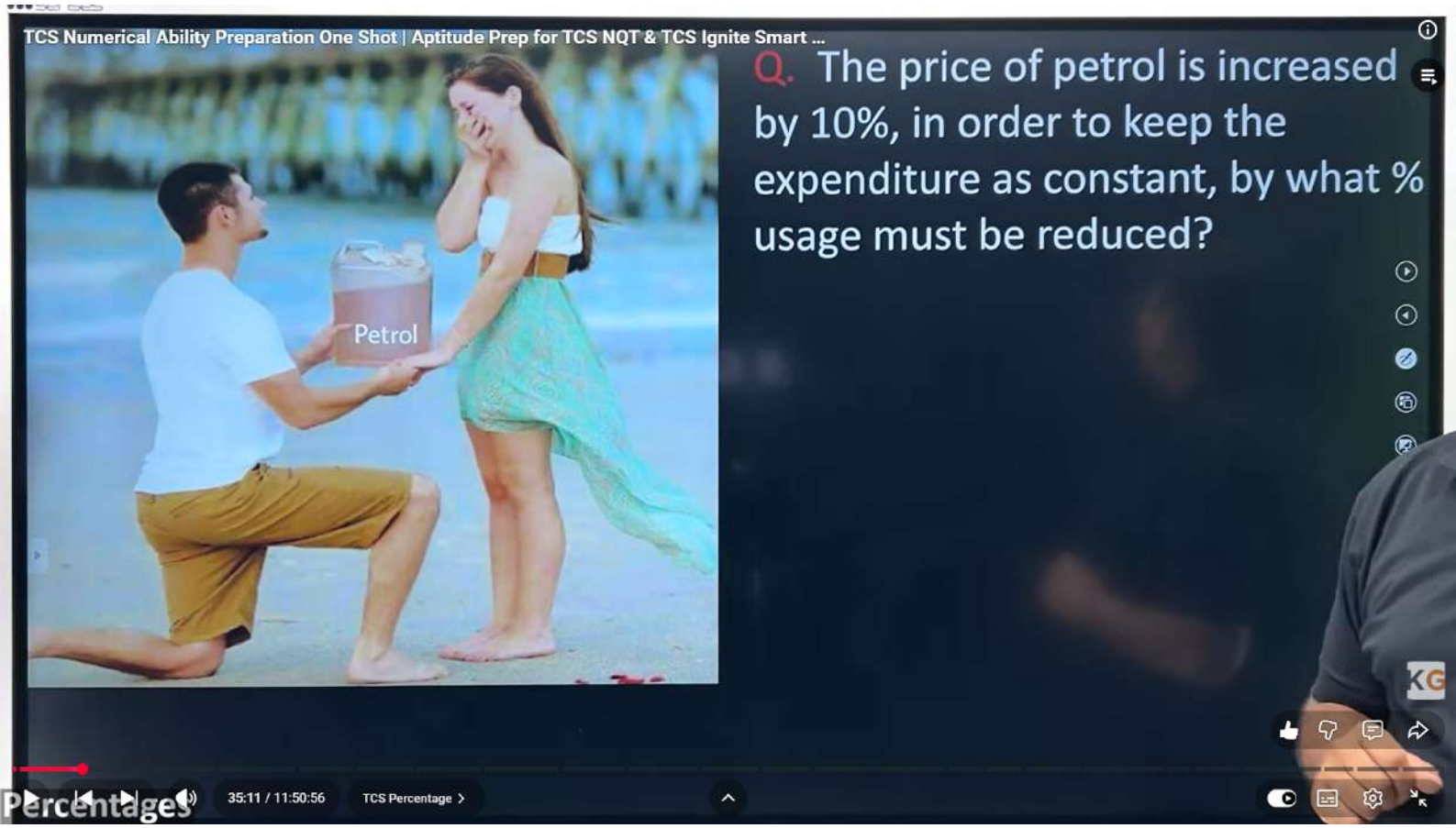
A. 14400

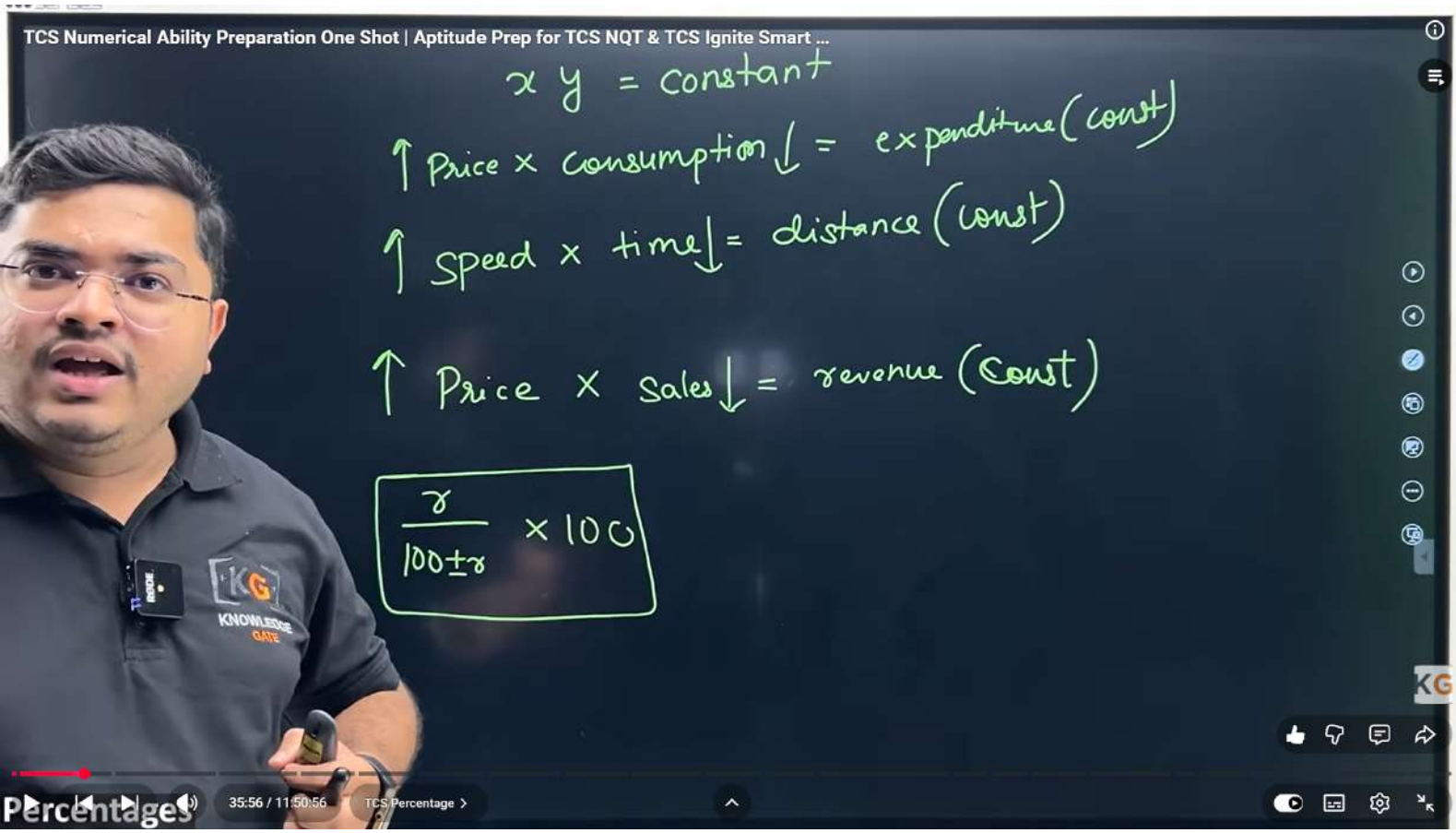
B. 4599

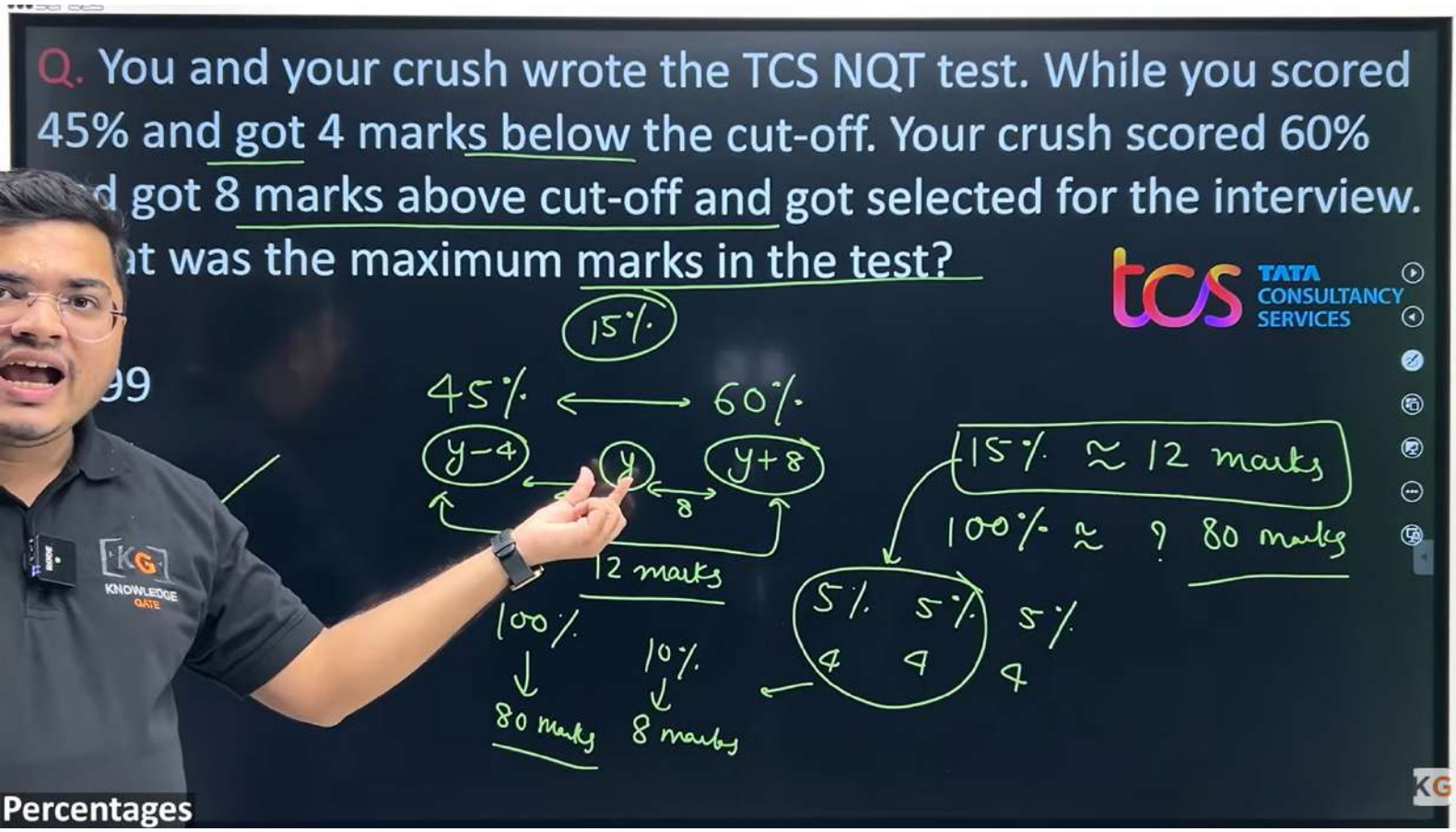
C. 59998

D. 23488

33:26 / 11:50:56 TCS Percentage


TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
Petrol
Q. The price of petrol is increased by 10%, in order to keep the expenditure as constant, by what % usage must be reduced?
35:11 / 11:50:56
TCS Percentage
Percentages


TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
x y = constant
↑ Price × consumption ↓ = expenditure (const)
↑ speed × time ↓ = distance (const)
↑ Price × sales ↓ = revenue (const)
r / (100 ± r) × 100
KNOWLEDGE GATE
35:56 / 11:50:56
TCS Percentage
Percentages


Q. You and your crush wrote the TCS NQT test. While you scored 45% and got 4 marks below the cut-off. Your crush scored 60% d got 8 marks above cut-off and got selected for the interview. t was the maximum marks in the test?
TATA CONSULTANCY SERVICES
15%
45% 60%
y−4 y y+8
8
12 marks
100%
80 marks
10%
8 marks
15% ≈ 12 marks
100% ≈ ? 80 marks
5% 5% 5%
4 4 4
KNOWLEDGE GATE
Percentages

Mean Mode & Median

Given data : 5 5 7 6 5

Mean / Average / Arithmetic Mean $= \dfrac{\text{Sum of all observations}}{\text{no. of observations}}$

$= \dfrac{5+5+7+6+5}{5}$

$= \dfrac{28}{5} = 5.6$

min
max

5 – 7

M Mode & Median

Gi data : 5 5 7 6 5

Median = central term [Median = 7 ✗]

asc / desc ⇒ 5 5 5 6 7

→ median

Data : 5 5 7 6 6 5

asc ⇒ 5 5 5 6 6 7

→ median = mid point / average

$\frac{5+6}{2} = \frac{11}{2}$

$= 5.5$

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Mean Mode & Median

Given data : 5 5 7 6 5

For 'n' terms arranged in ascending / descending order

(i) $n \rightarrow$ odd median $= \left(\frac{n+1}{2}\right)^{th}$ term

(ii) $n \rightarrow$ even median $= \dfrac{\frac{n}{2}^{th} + \left(\frac{n}{2}+1\right)^{th} \text{ term}}{2}$

Data: 5 5 5 6 7

$n = 5$ (odd)

median $= \frac{5+1}{2} = 3^{rd}$ term

Data: 5 5 5 6 6 7

$n = 6$ (even)

median $= \dfrac{\frac{6}{2}^{th} + \left(\frac{6}{2}+1\right)^{th} \text{ term}}{2}$

$= \dfrac{3^{rd} + 4^{th} \text{ term}}{2} = \frac{5+6}{2}$

Elementary Statistics
55:41 / 1:50:56
TCS Elementary Statistics

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Mean Mode & Median

Given data : 5 5 7 6 5

Mode ⇒ term with most occurances

or

term having highest frequency

$\boxed{\text{mode} = 5}$

Elementary Statistics

58:03 / 1:1:50:56

TCS Elementary Statistics